CHAPTER 3 SOME IMPORTANT PARTS REPLACEMENT **3-1**

3.1 GENERAL: 3-2
3.1.1 Attention *3-2*
3.1.2 List for tools in common *3-2*
3.2 BEARING BLOCK REPLACEMENT 3-5
3.2.1 Preparation *3-5*
3.2.2 Replacement procedure *3-6*
3.3 MOTOR COUPLING REPLACEMENT 3-7
3.3.1 Preparation *3-7*
3.3.2 Replacement procedure *3-8*
3.4 DRUM COUPLING REPLACEMENT 3-10
3.4.1 Preparation *3-10*
3.4.2 Replacement procedure *3-10*
3.5 MOTOR BRAKE REPLACEMENT 3-13
3.5.1 Preparation *3-13*
3.5.2 Replacement procedure *3-14*
3.5.2.1 A whole brake replacement 3-14
3.5.2.2 Thruster replacement 3-15
3.5.2.3 Spring square tube replacement 3-15
3.5.2.4 Spring replacement 3-16
3.5.2.5 Auto compensate device replacement 3-16
3.5.2.6 Brake gasket replacement 3-17
3.5.2.7 Hand release device replacement 3-18
3.5.2.8 Equalization nut replacement 3-18
3.6 DRUM BRAKE REPLACEMENT 3-19
3.6.1 Preparation *3-19*
3.6.2 Replacement procedure *3-19*
3.7 WHEEL BRAKE REPLACEMENT 3-19
3.7.1 Preparation *3-19*
3.7.2 Replacement procedure *3-20*
3.7.2.1 The whole wheel brake replacement 3-20
3.7.2.2 Hydraulic cylinder replacement 3-20
3.7.2.3 Brake gasket replacement 3-21
3.7.2.4 Block spring replacement 3-22
3.7.2.5 Block replacement 3-22
3.8 GANTRY WHEEL REPLACEMENT 3-23
3.8.1 Preparation *3-23*
3.8.2 Replacement procedure *3-23*
3.9 TROLLEY WHEEL REPLACEMENT 3-25
3.9.1 Preparation *3-25*
3.9.2 Replacement procedure *3-26*
3.10 MAIN HOIST PULLEY REPLACEMENT 3-28
3.10.1 Preparation *3-28*
3.10.2 Replacement procedure *3-28*

Chapter 3 Some important parts replacement

Summary：

Inevitably, there are so many processes such as abrasion, fatigue, deformation and ageing in QC life. Proper operation and maintenance in time will help parts on the expected service life but when the end coming, how to renovate them is very important in QC modernization management.

Unit assemblage replacement，it means replace the spoiled parts by new or repaired one. Mostly, the QC down time for restoration will decrease obviously by using the parts prepared previously, and then the economic losses will be reduced to a minimum

In this chapter, we will illustrate **replacement procedure** for QC main components which need disassembly and disengagement by figure method.

3.1 General:

3.1.1 Attention

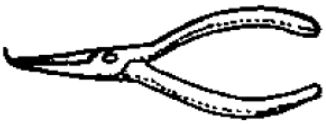
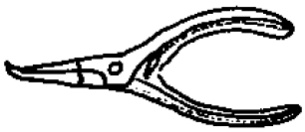
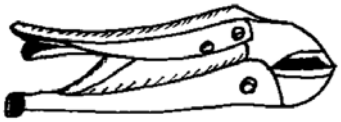
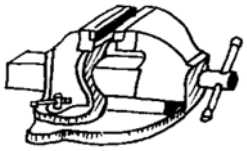
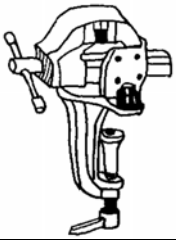
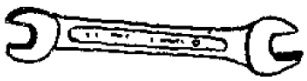
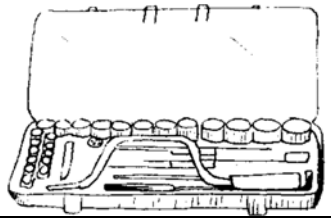
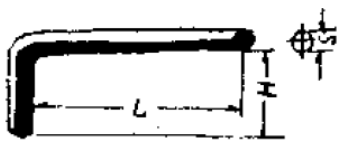
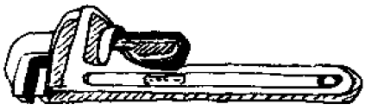
This chapter includes main components such as: bearing block、motor coupling、drum coupling、motor brake、drum brake、gantry wheel brake、trolley wheel、gantry wheel 、main hoist sheave、rope idler roller and so on. The personnel who takes part in the replacement procedure must be acquainted with the structure of components.， control the operation step and have an ability of coping with the abrupt affairs. Before working, it is obligatory that set down the ethnics in detail according to this chapter, have a pre-work meeting, adopt to safety precautions， point someone who takes responsible for safe coordination, and strictly obeys the safety rule request in chapter 2.

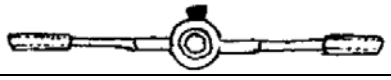
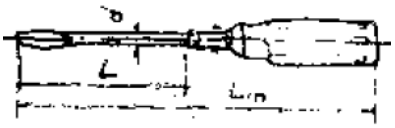
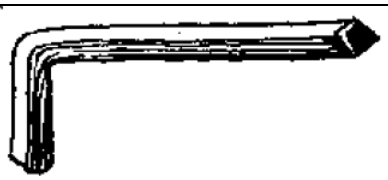
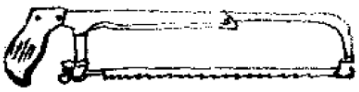
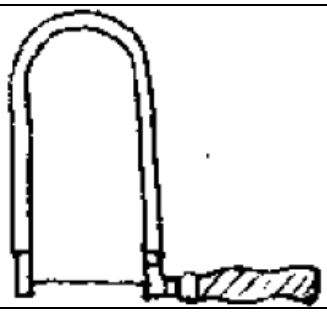
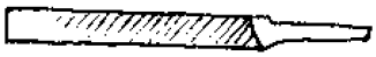
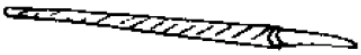
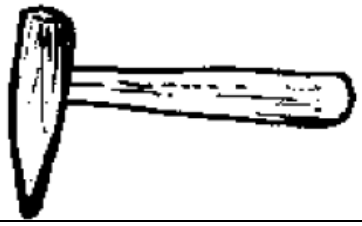
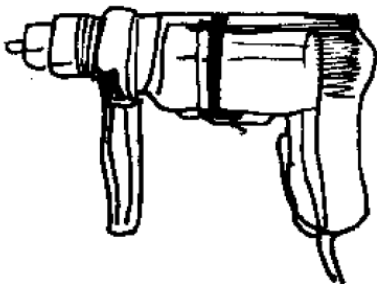
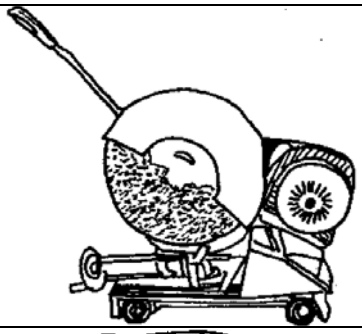
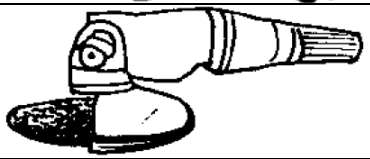
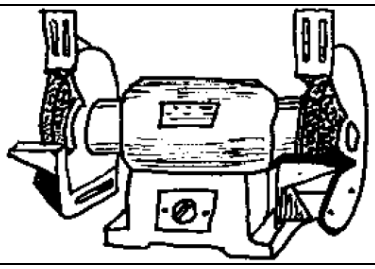
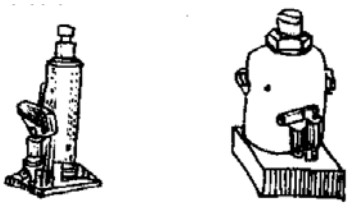
	Danger of tripping!
	Keep away from suspended loads!
	Pinch Hazard!

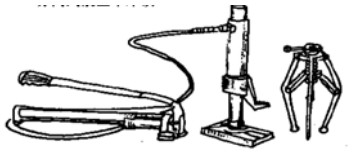
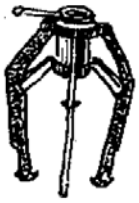
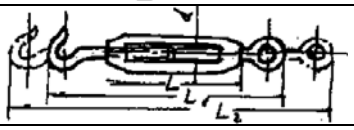
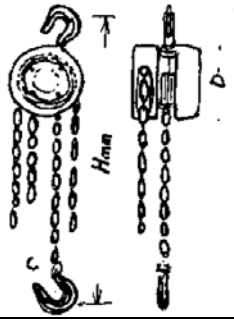
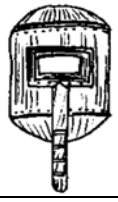
3.1.2 List for tools in common

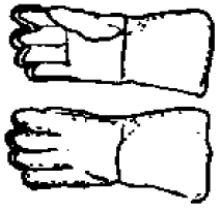
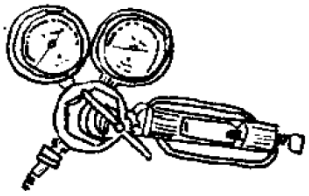
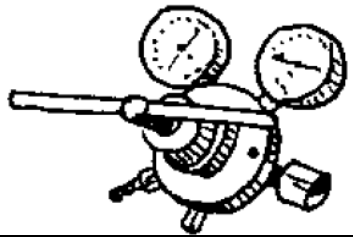
table 3-1 List some tools in common for replacement, yet only for reference.

table 3-1：List for tools in common

Wire pliers		Slip joint pliers	
Sharp nose pliers		Retainer pliers on hole	
Retainer pliers on shaft		Locking pliers	
Anvil vice		Table vice	
Double end wrench		Combination wrenches	
Offset ring spanner		Box spanner	
Inner hexagon spanner		Tension wrench	
Ratchet spanner		Pipe spanner	

Round die spanner		Tap wrench	
Bull screwdriver		Cross screwdriver	
Handle Inner hexagon screwdriver		Hack saw	
Jack saw		Scroll saw	
Blunt file		Blunt round file	
Fitter’s hammer		Chisel	
Electric hand drill		Electric sectional material cutter	
Orthotropic electric grinder		Angular electric abrader	
Floor stand grinder		Hydraulic jack	

Jacking screw		Separable hydraulic jack	
Separable hydraulic jack accessory（larma）		Additional pipe	
Open end hook block sheave		Open end chain block sheave	
Closed end hook block sheave		Closed end chain block sheave	
Rigging collar		Rigging shackle	
Rigging spiral flute		Rigging spiral flute（CO type）	
Rope cramp		Loop chain handle bottle gourd	
Welding pliers		Electric welder’s helmet	

Arc welder's gloves		Welder's boot	
Welding torch		Cutting torch	
Acetylene reducer		Oxygen bottle	
Oxygen reducer		Handle arc welding electrode	（1） Acid electrode （2） Basic electrode

3.2 Bearing block replacement

3.2.1 Preparation

The bearing blocks are included in main hoist mechanism, trolley driven mechanism, boom hoist mechanism, which are all level form.，shown in figure 3-2. Before replacement, carefully research the drawings, understand the bearings constitutes，the drawings about bearing block on table 3-2. personnel and tools on **table 3-3**.

table 3-2：Drawings for bearing block replacement

Main hoist bearing block	J526A0300，J526A0311
Trolley bearing block	J526A0400，J526A0404
Boom hoist bearing block	J526A0500，J526A0502

table 3-3：personnel and tools for bearing block replacement

Personnel/quantity		Tools/quantity	
Lifting worker	3	Bottle gourd	1（10t）
welder	1	larma	1
fitter	1	Tension wrench	2
electrician	1	others	several

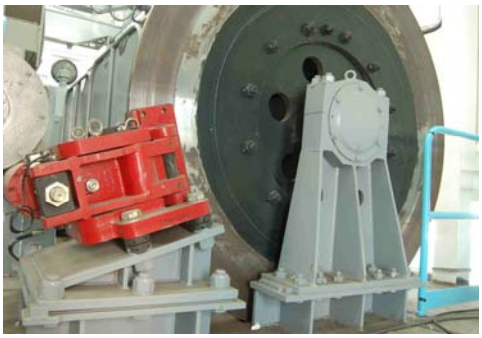


figure 3-1：Bearing block appearance

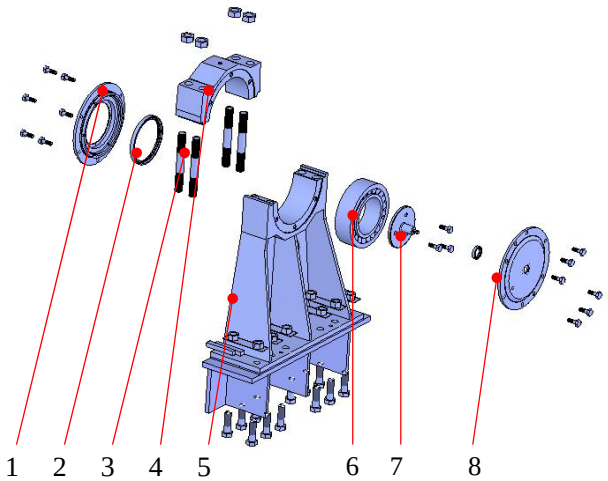


figure 3-2：Bearing block sketch map

1end over1　2 seal　3double screw bolt　4upper seat
5down seat　6 bearing　7L.S shaft　8 end cover2

3.2.2 Replacement procedure

1) Lower the spreader to the ground and slacken the Main Hoist ropes. (or position the boom to the level and slacken the Boom Hoist ropes)

2) Lock out the crane's motion for the affected drives by performing lock-out and tag-out procedure.

3) Disconnect the coupling between the drum and the rotary limit switch, Use slings and softeners to connect the drum to the service crane.

4) Tie off each of the ropes in machinery house to remove the rope load from the drum.

5) Slacken enough rope, so that the drum can be moved slightly. Before replacement the whole appearance of mechanism are shown in figure 3-3.

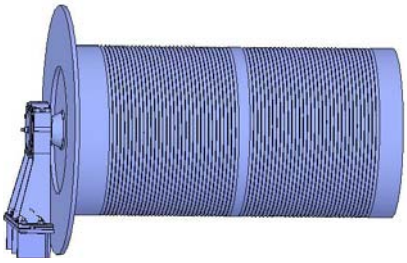


figure 3-3： The whole appearance of mechanism before replacement

6) Remove the bearing block end cover nuts.



figure 3-4：Step 6

7) Unfasten the stud nuts and screw.

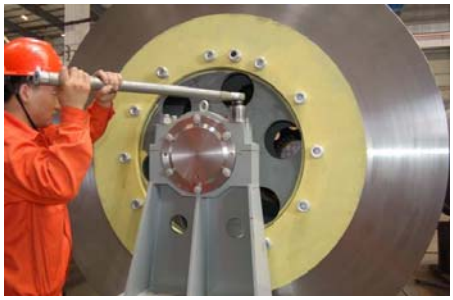


figure 3-5：Step 7

8) Remove the bearing block end cover 1



figure 3-6：Step 8

9) Remove the L.S shaft or shaft end plate



figure 3-7：Step 9



figure 3-8：Step 9

	The drum end is fasten on the support in machine house by ropes. Lift the drum up 50mm by bottle gourd, then move the lower bearing block sideways from the drum shaft by the service crane，use a puller with a 10-ton capacity to pull out of the bearing.

10) Carry the bearing block to the ground by the service crane, other procedure will do at workshop.



figure 3-9：Step10



figure 3-10：Appearance after up block removed

11) Remove the bearing and end cover 2.



figure 3-11：Step 11

12) Clean the drum shaft and bearing block with solvent. Shaft seals should also be replaced if necessary.



figure 3-12：Step 12

13) Install a new bearing.



figure 3-13：Step 13

14) Re-assemble the bearing block together in the reverse order mentioned above

15) Realign the drum couplings. Refer to chapter 2 for alignment criteria. Verify the drum-mounted rotary limit switch setting and alignment.

16) Remove LOTO devices and Pre-Op the crane. Startup and monitor the crane to verify its operation before returning it to service.

3.3 Motor coupling replacement

3.3.1 Preparation

The motor couplings are included in main hoist mechanism, trolley driven mechanism, boom hoist mechanism，which all are elastic coupling，shown in figure 3-15. Before replacement, carefully research the drawings, understand the motor coupling constitutes，the drawings about motor coupling on .table 3-4, personnel and tools on table 3-5.

table 3-4： Drawings for motor coupling replacement

Main hoist motor coupling	J526A0300
Trolley driven motor coupling	J526A0400
Boom hoist motor coupling	J526A0500

table 3-5：Personnel and tools for motor coupling replacement

personnel/ quantity		tools/quantity	
Lifting worker	1	Bottle gourd	1（10t）
welder	1	larma	1
fitter	3	Torque wrench	2
electrician	1	others	several



figure 3-14：Motor coupling appearance

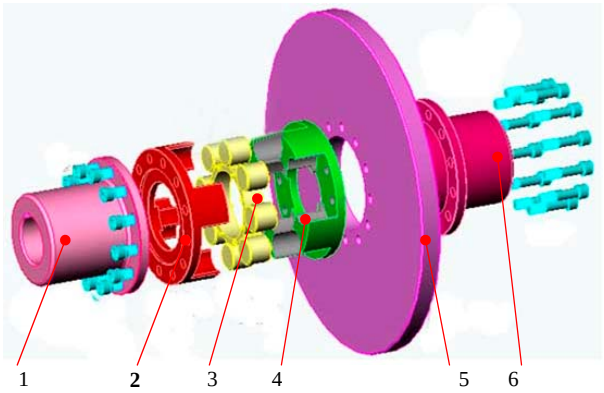


figure 3-15：Motor coupling sketch map

1	motor side coupling	2	motor side jaw
3	spider	4	reducer side jaw
5	brake disc	6	reducer side coupling

3.3.2 Replacement procedure

1) Lower the spreader to the ground and slacken the Main Hoist ropes. (or position the boom to the level and slacken the Boom Hoist ropes)

2) Lock out the crane's motion for the affected drives by performing lock-out and tag-out procedure.

3) Disconnect the coupling between the drum and the rotary limit switch, Use slings and softeners to connect the drum to the service crane.

4) Tie off each of the ropes in machinery house to remove the rope load from the drum.

5) Remove the electrical connection from the motor brake, release the brake manually, then remove the brake by service crane.

6) Unfasten the nuts between motor side half coupling and motor side jaw; remove the motor and half coupling by service crane.



figure 3-16：Step 6

7) Take out the motor side jaw.



figure 3-17：Step 7

8) Take out the spider.

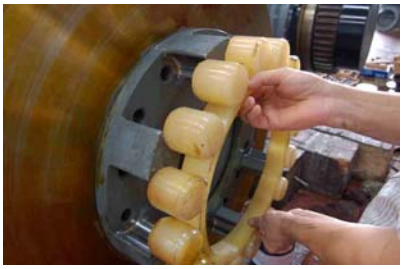


figure 3-18：Step 8

9) Remove the motor side half coupling by larma. Protect the motor shaft during the process.



figure 3-19：Step 9

10) Remove the reducer side half coupling. At first, remove the chain pulley on it.



figure 3-20：Step 10

11) Generally ， the chain pulley is two half forms. Remove it by service crane.

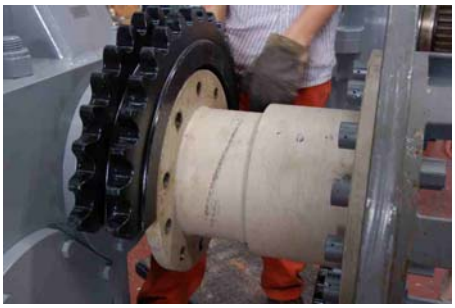


figure 3-21：Step 11

12) Remove the driven side jaw.



figure 3-22：Step 12

13) Remove the brake disc.

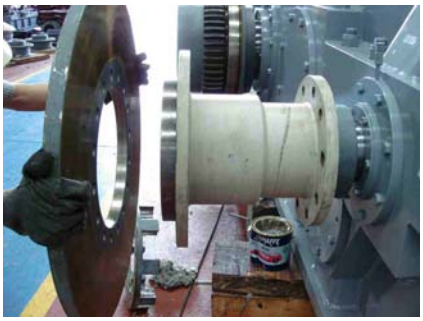


figure 3-23：Step 13

14) Remove the driven side half coupling by larma，Protect the reducer shaft during the process.



figure 3-24：Step 14

15) Next is the procedure of re-align the new motor coupling. A bought coupling should be anti-corrosive treated, sealed well.



figure 3-25：Appearance of folded motor coupling

16) After unfold the cover, clean the anti-rust oil, heating inset the two half couplings(or by oil bath heating)



figure 3-26：Step 16

17) When re-aligning, match the serial number between the half coupling and the brake disc.



figure 3-27：Step 17

18) Thread the heating half coupling into the, reducer shaft for blocking proof, the quicker the better, not overtime(ten minute).



figure 3-28：Step 18

	Normally, the serial number in disc includes item number, mechanism number, and brake disc front or back (A or B). And A aims at half coupling, B aims at half flange.

19) Realign the motor coupling according to the method in chapter 2, as for coupling allowable compensation rate, install separation, nuts torque, please refer to the coupling manual.

3.4 Drum coupling replacement

3.4.1 Preparation

The drum couplings includes main hoist mechanism, trolley driven mechanism, boom hoist mechanism，which all are dental formula coupling, shown in **figure 3-30**. Before replacement, carefully research the drawings, understand the drum coupling constitutes，the drawings about drum coupling on table 3-6. personnel and tools on **table 3-7**.

table 3-6：Drawings for drum coupling replacement

Main hoist drum coupling	J526A0300
Trolley driven drum coupling	J526A0400
Boom hoist drum coupling	J526A0500

table 3-7：Personnel and tools for drum coupling replacement

Personnel/quantity		Tools /quantity	
Lifting worker	2	Bottle gourd	2（10t）
painter	1	larma	1（200t）
locksmith	3	Torque wrench	2
electrician	1	others	several

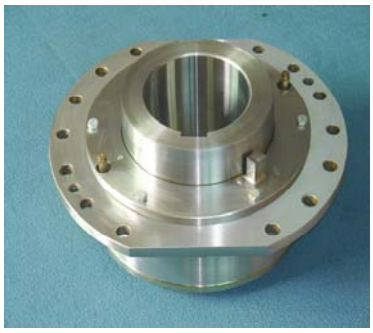


figure 3-29：Drum coupling appearance

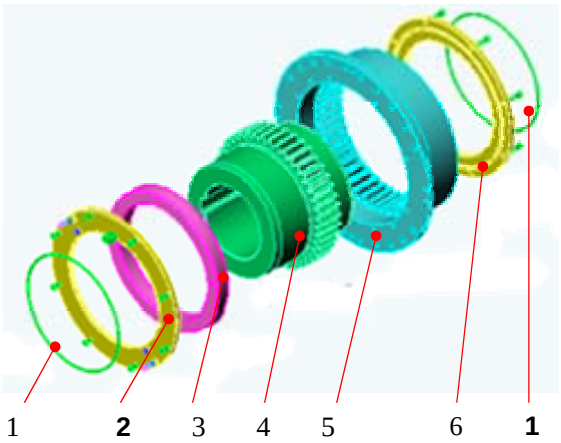


figure 3-30：Drum coupling sketch map

1 seal 2 out cover 3 load ring
4 out dental boss 5 inner dental flange 6 inner cover

3.4.2 Replacement procedure

1) Lower the spreader to the ground and slacken the Main Hoist ropes. (or position the boom to the level and slacken the Boom Hoist ropes)

2) Lock out the crane's motion for the affected drives by performing lock-out and tag-out procedure.

3) Disconnect the coupling between the drum and

the rotary limit switch, Use slings and softeners to connect the drum to the service crane.

4) Bind each of the ropes to machinery house to remove the rope load from the drum.

5) Hang the drum by service crane, remove the bearing block, brake disc, and drop out the drum from the reducer shaft.

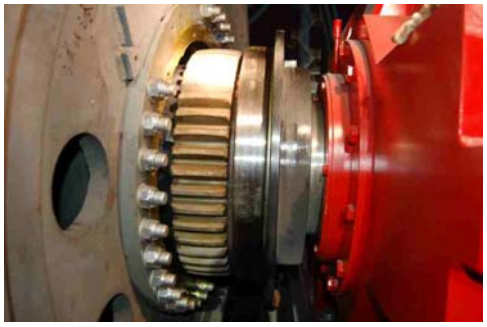


figure 3-31：Step 5

6) Disjoin the drum coupling step by step, and protect the out dental boss and inner dental flange.

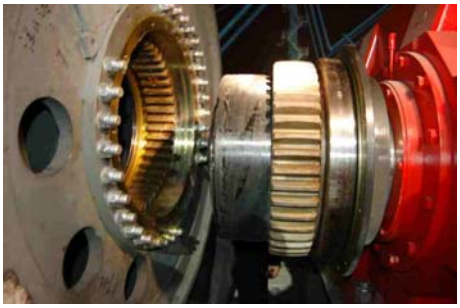


figure 3-32：Step 6



figure 3-33：Demounted inner dental flange



figure 3-34：Demounted out dental boss

7) Then we will introduce how to demount the out dental boss and inner dental flange separately. For inner dental flange, at first place the drum in area near assembly. Then unfasten the nuts on it.



figure 3-35：Step 7

8) Suspended the inner dental flange by service crane，drop out it from the drum step by step.



figure 3-36：Step 8

9) Before demounting the out dental boss, at first estimate it if it is useful or not, If it is to be discarded as useless, plane it from the key slot by fire, then the coupling will drop out easily.



figure 3-37：Step 9

10) During the process, the reducer also can be set down (shaft upraise vertical), it is benefit to jacking up. Then fasten the double screw nut into the out dental boss.



figure 3-38：Step 10

11) Add an appropriate high mat slice，In order to make use of the shims of the jack to rise height well，(note all the shims should be stuffed).



figure 3-39：Step 11

12) Put the jack on the top of the shims steadily.



figure 3-40：Step 12

13) Put the raising flange at the peak, and make the double screw through the hole in flange..



figure 3-41：Step 13

14) Tighten nuts, assure the interface level up between the raising flange and jack, completely contact.



figure 3-42：Step 14

15) Start-up the jack,. Normally, the whole procedure need one hour, more or less.



figure 3-43：Step 15

16) Next is the procedure of realign the new drum coupling. A bought coupling should be anti-corrosive treated, sealed well.



figure 3-44：Appearance of folded drum coupling

17) For cleaning rust inhibiting oil,benefit to next step, at first ,take apart the inner dental flange, outer dental boss and abrasion indicator



figure 3-45：Step 17

18) Clean the inner dental flange and outer dental boss carefully.



figure 3-46：Step 18

19) Heating inset the out dental boss,(or by oil bath heating)



figure 3-47：Step 19

20) When preparation ends, start up the installation, at first, telescope the abrasion indicator.



figure 3-48：Step 20

21) Telescope the out dental boss at the right place where is on the shaft of reducer, the quicker the better, for it will shrink in short time.



figure 3-49：Step 21

22) After the out dental boss fixed, telescope the abrasion indicator.



figure 3-50：Step 22

23) Realign the drum coupling according to the method in chapter 2, as for coupling allowable compensation rate, install separation, please refer to the coupling manual.

24) Remark line at the abrasion indicator, according to chapter 2/

3.5 Motor brake replacement

3.5.1 Preparation

The motor brake includes main hoist mechanism, trolley driven mechanism, boom hoist mechanism，which are all electric hydraulic disc brakes., shown in figure 3-52, Before replacement, carefully research the drawings, understand the motor brake constitutes，the drawings

about motor brake on table 3-8, the related personnel and tools on table 3-9.

table 3-8： Drawings for motor brake replacement

Main hoist motor brake	J526A0300
Trolley driven motor brake	J526A0400
Boom hoist motor brake	J526A0500

table 3-9：Personnel and tools for motor brake replacement

Personnel / quantity		Tools /quantity	
Lifting worker	1	bottle gourd	1（5t）
welder	1	Fitter tools	several
locksmith	2	Torque wrench	1
electrician	1		



figure 3-51：Motor brake appearance

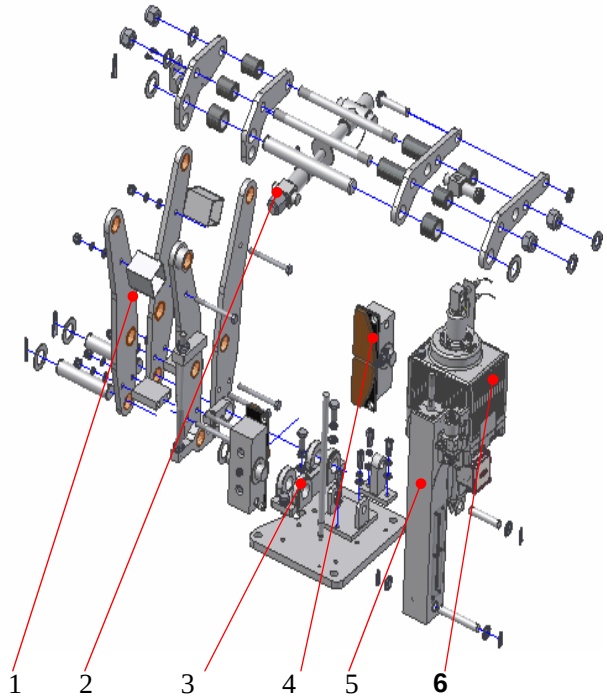


figure 3-52： motor brake sketch map

1 brake arm 2brake draw bar 3 equalization nut
4 brake gasket 5 brake spring subassembly 6 thruster

3.5.2 Replacement procedure

1) Lower the spreader to the ground and slacken the Main Hoist ropes. (position the boom to the level and slacken the Boom Hoist ropes).

2) Avoid the crane's motions which affect save driving by performing lock-out and tag-out procedure.

3) Only it is necessary，remove the whole brake., next we will introduce how to replace the whole brake and its ass.

3.5.2.1 A whole brake replacement

4) Remove the power wire from the thruster, remove the L.S, and release the brake by hand.

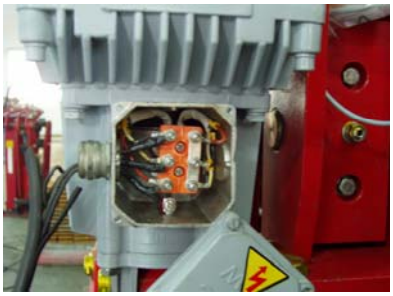


figure 3-53：Step 4

5) Remove the nuts between the brake and support.



figure 3-54：Step 5

6) Suspend the brake away, mark the original position.



figure 3-55：Step 6

7) According to the mark, full install a new brake, relocation and adjustion is needed if necessary.



figure 3-56：Step 7

8) Remove LOTO devices and Pre-Op the crane. Startup and monitor the crane to verify its operation before returning it to service.

3.5.2.2 Thruster replacement

1) Take out the splint pin from up and down shaft on thruster.



figure 3-57：Step 1

2) Pull out the pin roll on thruster end, bend the thruster for laid aside firmly.



figure 3-58：Step 2

3) Pull out the pin roll on the support.



figure 3-59：Step 3

4) Suspend the thruster away. Thruster internal structure is more complicated, don not repair it in site. If it has any fault, please remove the whole thruster and give it to the professional person for repairing.

5) Install a new thruster.

3.5.2.3 Spring square tube replacement

1) At first, pull out the pin roll on thruster end,, bend the thruster for laid aside firmly.



figure 3-60：Step 1

2) Turn the torque adjustable double screw bolt counter-clockwise，release the spring force.



figure 3-61：Step 2

3) Pull out the elastic cylindrical pin from groove nuts,. Remove the groove nuts on square tube.



figure 3-62：Step 3

4) Pull out the splint pin from the spring support.



figure 3-63：Step 4

5) Pull out the pin roll.



figure 3-64：Step 5

6) Put up the set square.



figure 3-65：Step 6

7) Pull out the spring square tube.



figure 3-66：Step 7

3.5.2.4 Spring replacement

1) Screw the torque adjustable screw bolt to nut



figure 3-67：Step 1

2) Turn the torque adjustable screw bolt clockwise, make sure that the screw bolt doesn't hold back the dummy plate under tube.



figure 3-68：Step 2

3) Pull out the dummy plate under spring tube by appropriate hammer.



figure 3-69：Step 3

4) Turn the torque adjustable screw bolt counter clockwise，until spring force disappear absolutely，wrest the nuts on the bolt up end. Pull out the bolt, spring support and spring. Install a new spring.



figure 3-70：Step 4

3.5.2.5 Auto compensate device replacement

1) Remove the compensate plate..



figure 3-71：Step 1

2) Remove the splint pin on shaft, loosen the holding screw on the pin roll/



figure 3-72：Step 2

3) Pull out the compensate screw bolt from cross square tube.



figure 3-73：Step 3

4) Loosen the nuts on screw bolt.



figure 3-74：Step 4

5) Pull out the screw shaft.



figure 3-75：Step 5

6) Pull out the pin roll.



figure 3-76：Step 6

7) Pull out the auto compensate device and set square assembly from the brake arm.



figure 3-77：Step 7

8) Remove the set square, take out the auto compensate device, install a new one.



figure 3-78：Step 8

3.5.2.6 Brake gasket replacement

1) Loosen the brake release device manually. Revolve the adjustable nut sufficiently.



figure 3-79：Step 1

2) Loosen the gasket binding bolt.



figure 3-80：Step 2

3) Move the gasket level for breaking away key. Then pull out it, and re-align a new gasket.



figure 3-81：Step 3

3.5.2.7 Hand release device replacement

1) Loosen the holding bolt on hand grip



figure 3-82：Step 1

2) Pull out hand release grip.



figure 3-83：Step 2

3) Loosen the holding bolt on idler wheel.



figure 3-84：Step 3

4) Pull out the idler wheel.



figure 3-85：Step 4

3.5.2.8 Equalization nut replacement

1) Loosen the nuts under the screw bolt.



figure 3-86：Step 1

2) Loosen the adjustable nut to the screw end.



figure 3-87：Step 2

3) Loosen the adjustable screw bolt to the screw end. and the top nuts.



figure 3-88：Step 3

4) Pull out the screw bolt, install a new one.



figure 3-89：Step 4

3.6 Drum brake replacement

3.6.1 Preparation

The drum brake are included in main hoist mechanism, boom hoist mechanism，which are all hydraulic disc brake, shown in , Before replacement, research the drawings carefully, understand the drum brake constitutes，the drawings about drum brake on table 3-10, related personnel and tools on table 3-11.

table 3-10：Drawings for drum brake replacement

Main hoist drum brake	J526A0300
Boom hoist drum brake	J526A0500

table 3-11：Personnel and tools for drum brake replacement

Personnel/quantity		Tools /quantity	
Lifting worker	1	Bottle gourd	1（5t）
Welder	1	Fitter tools	several
locksmith	2	Torque wrench	1
electrician	1		



figure 3-90：Drum brake appearance

3.6.2 Replacement procedure

1) Lower the spreader to the ground and slacken the Main Hoist ropes. (If change the drum brakes of the boom hoist mechanism, please position the boom to the level and slacken the Boom Hoist ropes)。

2) Lock out the crane's motions which affect save driving by performing lock-out and tag-out procedure.

3) If it is necessary，remove the whole brake., next we will introduce how to replace the whole brake and its ass.

For **Details , please refer to brake manual.**

3.7 Wheel brake replacement

3.7.1 Preparation

The wheel brakes are included in the gantry mechanism, which are hydraulic disc brakes, see figure 3-92. Before replacement, it's very important to read the drawings on the wheel brake's structure. The drawings about wheel brakes see table 3-12, Personnel and tools placement see table 3-13.

table 3-12：The drawings on wheel brakes

Wheel brakes	J526A0600，J526A0607

table 3-13：Personnel and tools

personnel/ quantity		tools/quantity	
Lifting worker	1	Handle gourd	1（5t）
welder	1	Fitter tools	several
fitter）	2	Torque spanner	1
electrician	1		



figure 3-91：Wheel brake natural figure

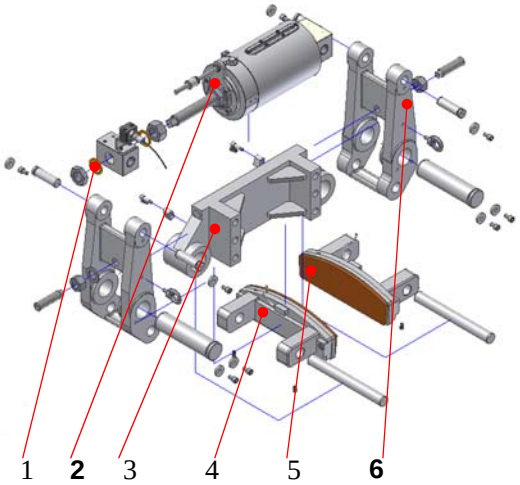


figure 3-92：Wheel brake abridged view

1 on-off key	2 fluid cylinder	3 base
4 block	5 retaining piece	6 brake arm

3.7.2 Replacement procedure

Only on special conditions, the whole wheel brake should be put apart. We introduce the replacement procedure of wheel brake and other unit mounts in the following parts.

3.7.2.1 The whole wheel brake replacement

1) Lock out the crane's motions which affect save driving by performing lock-out and tag-out procedure.
2) Remove the wire on the wheel brake L.S.
3) Hand releases the wheel brake through the hydraulic state.
4) Loosen the holding nuts, release the pressure in hydraulic state, loosen the adjustable nuts, and erase the spring force.



figure 3-93：Step 4

5) Remove the oil pipe on wheel brake, put the hose to a prepared case，warding for the hydraulic oil flowing to the ground.
6) Bracket the wheel brake by fork truck, and remove the bolts on its support.



figure 3-94：Step 6

7) Suspend the wheel brake by fork truck.
8) Install a new wheel brake by fork truck, connect the oil hose, and adjust it according to chapter 2.
9) Remove LOTO devices and Pre-Op the crane.
10) Startup and monitor the crane to verify its operation before returning it to service.

3.7.2.2 Hydraulic cylinder replacement

	Hydraulic cylinder is a main execute component on wheel brake, need special tools to assemble it, it is difficult to repair it in site. If there is failure in it, please replace the whole cylinder to professional person and repair it in workshop.

1) Lock out the crane's motion which affect save driving by performing lock-out and tag-out procedure
2) Disappear the spring force on wheel brake, and remove the wire and hose imitate last step.



figure 3-95：Step 2

3) Loosen the inner hexagon screw on brake arm using for holding the retainer ring.



figure 3-96：Step 3

4) Pull out the pin roll



figure 3-97：Step 4

5) Pull out the whole cylinder.



figure 3-98：Step 5

6) Install a new, same type hydraulic cylinder, and re-assemble the L.S wire and hydraulic hose.



figure 3-99：Step 6

7) Adjust the wheel brake according to chapter 2.



figure 3-100：Step 7

3.7.2.3 Brake gasket replacement

	When gasket thickness less than 2mm， you should replace the gasket.

1) Lock out the crane's motion for the affected drives by performing lock-out and tag-out procedure.

2) Open the brake to the most opening span.

3) Loosen the bolts on gasket.



figure 3-101：Step 3

4) Hold the gasket by hand, and sever it to the wheel side, make the gasket break away the key, and pull out it along the wheel radial.



figure 3-102：Step 4

5) Inspect the gasket surface clean or not，deform or not , and greasy dirt or not

6) Install a new, same type gasket along wheel radial, and make the gasket groove aimed at join key groove.

7) Screw down all the bolts.

8) Adjust the wheel brake according to chapter 2

9) Remove LOTO devices and Pre-Op the crane.

3.7.2.4 Block spring replacement

	Normally,block spring will not failure, if find some abnormal condition, maybe the spring fall off or others .

1) Loosen the holding screw.



figure 3-103：Step 1

2) Loosen the holding block, pull out the old spring, replace a new one and mount it firmly.



figure 3-104：Step 2

3.7.2.5 Block replacement

	Normally,block will not failure, if find some abnormal condition, replace it according following procedure.

1) Lock out the crane's motions which affect driving by performing lock-out and tag-out procedure.

2) Loosen the spring.

3) Eliminate the fluid sealant on up and down block holding screw hole.

4) Remove the holding pin.



figure 3-105：Step 4

5) Pull out the block screw.



figure 3-106：Step 5

6) Pull out the old block along the wheel radial.



figure 3-107：Step 6

7) Install a new same type block.

8) Install holding screws and wipe the fluid sealant.

9) Install the spring.

10) Remove LOTO devices and Pre-Op the crane.

11) According to chapter2, adjust the block.

3.8 Gantry wheel replacement

3.8.1 Preparation

Gantry wheels are included in gantry mechanism, 8 wheels per corner, 32 wheels in total.，hereinto, 16 driving, wheels， other 16 are driven wheels, gantry wheels constitute shown on **figure 3-108**, Before replacement, it's very important to read the drawings on the gantry wheel structure. The drawings about wheel brakes see **table 3-14**, Tools and personnel shown in **table 3-15**.

table 3-14：Drawings for gantry wheels replacement

Gantry wheels mechanism	J526A0600 , J526A0604 , J526A0605

table 3-15：Personnel and tools for gantry wheels replacement

personnel/quantity		Tools /quantity	
Lifting worker	3	Bottle gourd	1（5t）
Welder	1	Fitter tools	Several
Locksmith	2	Torque wrench	1
Electrician	1		



figure 3-108：Gantry wheels view

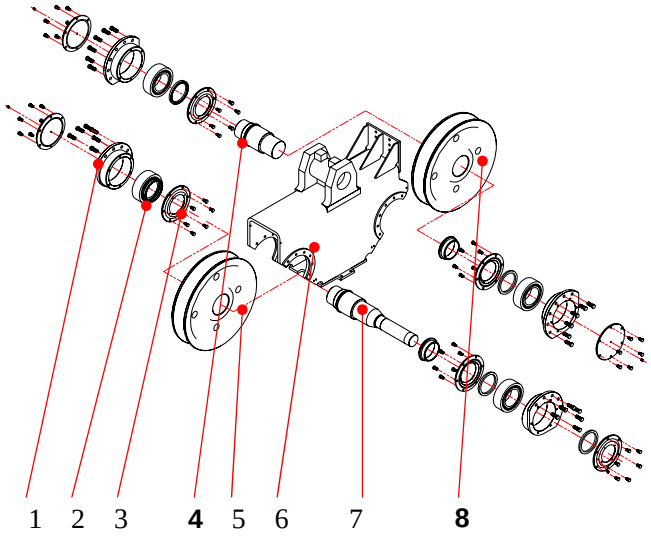


figure 3-109：Gantry wheels sketch map

1bearingblock 2 bearing 3through cover 4driven shaft
5driving wheel 6frame 7driving- shaft 8drivenshaft

3.8.2 Replacement procedure

1) Jack up the crane with no suspended load in calm weather.

2) Lock out the crane's motion for the affected drives by performing lock-out and tag-out procedure.

3) Move the trolley to the following position:

- At maximum outreach when the landside leg is jacked up.
- At parking position when the waterside leg is jacked up.

4) The boom position is:

- Raise the boom to the maintenance position when the waterside leg is jacked up.
- Set the boom to the operating position when the landside leg is jacked up.

At the jack point, put a –30x1000x1000mm steel plate cross the rail aligning with the center line of lower equalizer beam where the replacement is located at and a 250t hydraulic jack with at least 500mm strokes on this plate. The center line of jack should match the center line of rail. The wheel load value listed in chapter1.The jack capacity to jack up the lower equalizer beam is at least four times of the wheel load).

5) Align the hydraulic jack to center of lower equalizer beam. Put a -50X580X580 plate on top of the jack and let this plate contact with lower equalizer beam. Jack up the lower equalizer beam and have a forklift to hold the truck whose broken wheel is to be replaced. So there will be no load at truck pin.

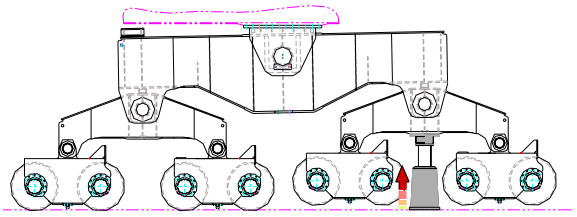


figure 3-110：Step 5

6) Jack up the equalizing beam by another jack, this procedure should match with big jack at the jack

point, until the clearance between shaft and hole is equal.



figure 3-111：Step 6

7) Push the shaft by hand.



figure 3-112：Step 7



figure 3-113：Step 7

8) Fall the bogie on the rail slowly, then translate it laterally, move the bogie to the workshop by fork truck.

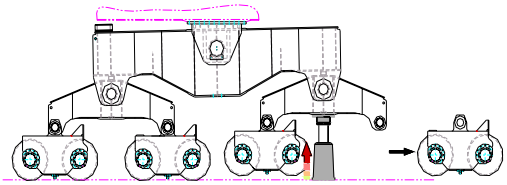


figure 3-114：Step 8

9) Replacement will be done in workshop, at first, remove the wheel end cover.



figure 3-115：Step 9

10) Remove the bolts on bearing block.



figure 3-116：Step 10



figure 3-117：Driving shaft apparent after step 10

11) Suspend away the wheel from the bogie by service crane.



figure 3-118：Step 11

12) Pull out the shaft from the bearing by larma. At

first, install additional pipe for larma on screw hole in bearing block.



figure 3-119：Step 12

13) Install the carrier rod of larma.



figure 3-120：Step 13

14) Install the hydraulic jack. Match the face between shaft end and jack. Then start up.

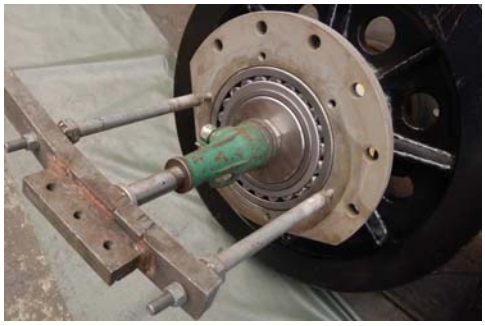


figure 3-121：Step 14

15) Pull out the bearing

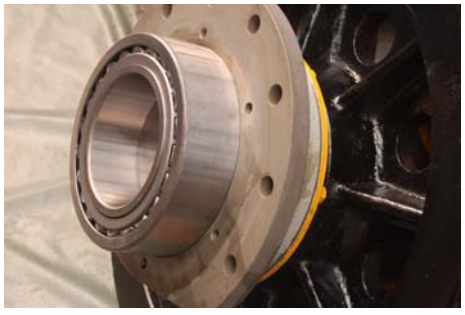


figure 3-122：Step 16

16) Pull out the bearing by hand.



figure 3-123：Step 16

17) Pull out the bearing block.

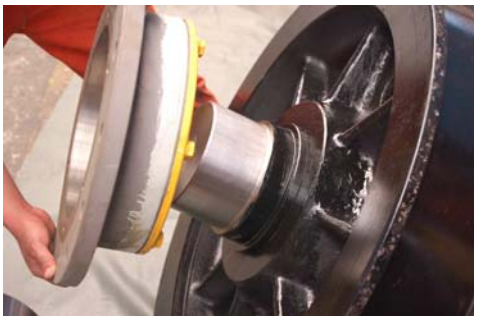


图 3-124：Step 7

18) Re-assemble the bearing and gantry wheel in contrary above on the gantry mechanism and equalizing beam.

3.9 Trolley wheel replacement

3.9.1 Preparation

Trolley wheels are included in the trolley ass, 2 wheels per corner, 8 wheels in total. For the pull type trolley, all the wheels are driven wheels. The wheels structure is shown in . Before replacement, it's very important to read the drawings on the trolley wheel's structure. The drawings about trolley wheel are shown in table 3-16, Personnel and tools placement are shown in table 3-17.

table 3-16：Drawings for trolley wheels replacement

Trolley wheel	J526A0200，ZT2-020936-00

table 3-17：Personnel and tools for trolley wheels Replacement

personnel/ quantity		Tools/quantity	
Lifting worker	3	Handle gourd	1（5t）
welder	1	Fitter tools	several
Locksmith	2	Torque spanner	1
Llectrician	1		

3.9.2 Replacement procedure

	On running, if there is noise or splashing metal dust from the wheels, you should stop at once and check it, make an adjustable scheme.
	The checking of New QC during running-in time（150～300h）, is very important. When built-in stress is released, trolley frame will deform appreciably, then check the trolley wheel and adjust it.
	On the end of adjusting, trail run with empty load, if well , then trail run with rated load.(20～30t, During this period, if there is anything wrong, adjust the wheel eccentric sleeve seat 2~5degree until acceptable.

Trolley wheel replacement requires considerable technology. Workers must read the ZPMC instructions carefully.

It is recommended that trolley wheel or axle repair work be accomplished in a workshop instead of on the crane.

Consider the following basic steps to accomplish the replacement:

1) Move the trolley to under the maintenance hole on the Machinery house deck.
2) Lift a new wheel assembly to the machinery house by service crane through the big hatch. Low the assembly to the trolley through the small hatch.
3) Move the trolley to the girder end under the outdoor service crane for maintenance.
4) Lower the spreader to the ground and slacken the main hoist ropes.
5) Lock out crane motion by proceeding lockout-tagout procedure.
6) Install a 10T jack under jacking pads which are on the four corner of the trolley. The gap between the top of rail and the bottom of jack point is about 365mm. Lift the trolley about 50 mm. The timber block may be necessary to put between under the trolley frame and rail top to prevent trolley from tilting.
7) The trolley wheel assembly is connected with a bolted connection to the trolley structure. The connections need to be unfastened to disengage the trolley wheel assembly from the trolley (**figure 3-141 section A-A and section B-B**).
8) Disconnect the defective wheel assembly and reassemble the new wheel ass. to the trolley.
9) Remove lock-out and tag-out devices and Pre-Op the crane.
10) Move the trolley to under the Machinery house.
11) Lift the old wheel assembly to machinery house by service crane from the small hatch, then low to the ground through the big hatch. (**figure 3-139)**

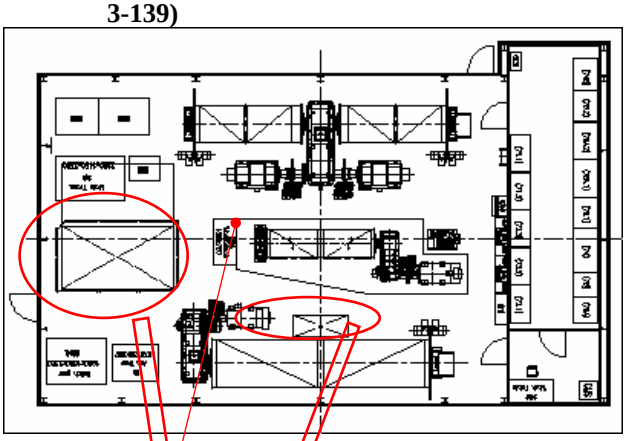


figure 3-139：M-house plan arrangement

12) Disassemble the replaced wheel assembly at the workshop. For detailed inside of trolley wheel assembly, please refer to the **figure 3-140** trolley wheel and axle assembly drawing.

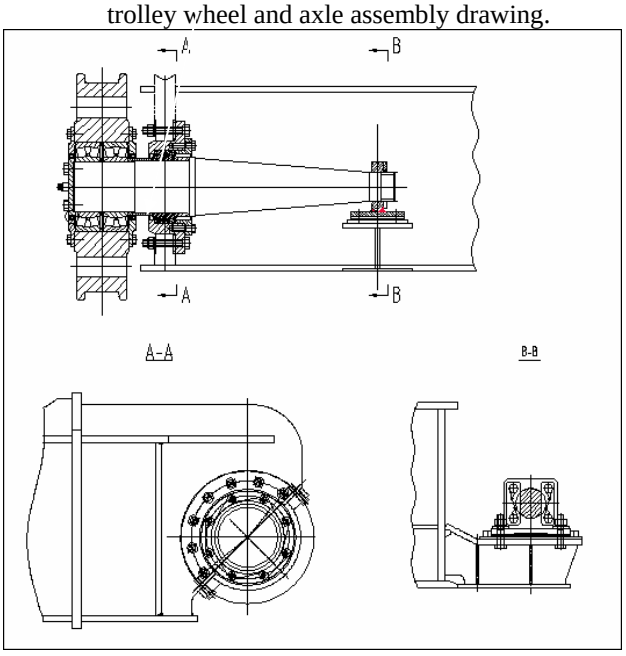


figure 3-140：Trolley Wheel and Axle Ass

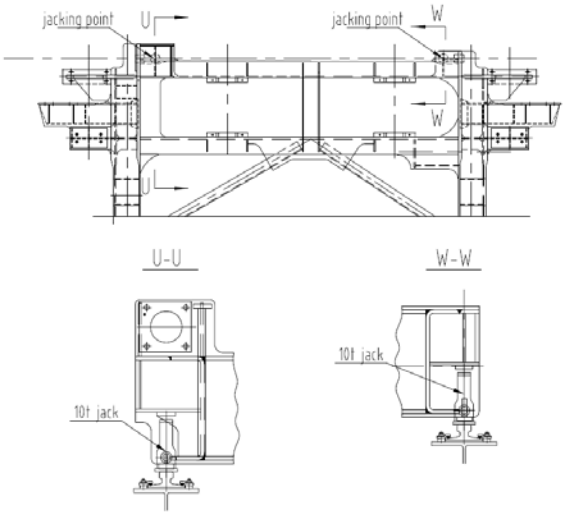


figure 3-141：Jacking point on the trolley

3.10 Main hoist pulley replacement

3.10.1 Preparation

The main hoist pulleys are included at boom end,, trolley frame, headblock、girder end, 40 in total. The nylon pulley at boom end needn't replaced, so the pulleys following are the forged one at other positions, 36 in total. The main hoist pulley pulleys constitutions are shown in **figure 3-**, before replacement, carefully research the drawings, understand the pulley constitutes，the drawings about pulley on **table 3-18**, personnel and tools on **table 3-19**.

table 3-18：Drawings for pulley replacement

Main hoist pulley at trolley	J526A0200 ， ZT2-021023-00 ， ZT2-021123-00
Main hoist pulley at headblock	J526A1200
Pulley in main hoist	J526A1000

table 3-19：Personnel and tools for pulley replacement

personnel/quantity		tools/quantity	
Lifting worker	3	Bottle gourd	2（0.5t）
Welder	1	Fitter tools	Several
Locksmith	2	Torque wrench	1
Electrician	1		



figure 3-125：Main hoist pulley appearance

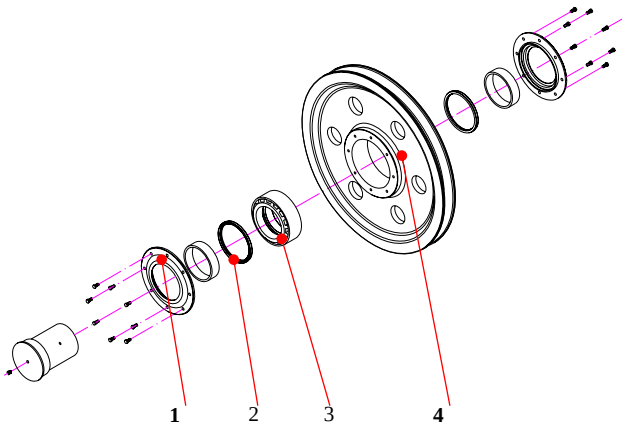


figure 3-126：Main hoist pulley sketch map

1 end cover　2 seals　3 bearing　4 pulley

3.10.2 Replacement procedure

1) Move the trolley to backreach so that the sheaves which are to be replaced are accessible with the bridge service crane at backreach.

2) Lower the headblock to the ground and slacken the main hoist ropes.

3) Lock out crane hoist and trolley motions by performing the lock-out and tag-out procedure.

4) Loosen the 8 bolts connected the shaft upper-sea.



figure 3-127：Step 4

5) Use two 0.5t chain hoists to remove the ropes from the sheaves which are to be replaced.

6) Remove the 2 shaft upper-seats and lift up the sheave assembly. Once the ropes are clear of the sheaves, place a timber across the opening where the sheaves were and released the rope wire onto the timber. Pull the bridge toward landside to let the sheave assembly be accessed by the electrical hoist on the jib.

7) Lower the sheave assembly to the ground and repair damaged sheave assembly parts in a workshop.



figure 3-128：Step 7

8) Tap out the shaft, don't damage the lubrication hole.



figure 3-129：Step 8

9) Suspend the shaft away by service crane in workshop.



figure 3-130：Step 9

10) Separate the two pulleys by service crane.



figure 3-131：Step 10



figure 3-132：Dummy plate between pulles

11) After unfastening the pulley, remove the bolts on trouble pulley end cover.



figure 3-133：Step 11

12) Pull out up half bearing by larma.



figure 3-134：Step 12



figure 3-135：Dummy plate between bearings

13) Pull out the dummy plates.



figure 3-136：Step 13

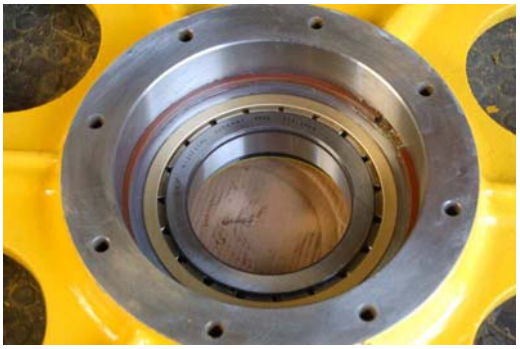


figure 3-137：outer dummy plate（brown part）

14) Turn over the pulley, pull out another bearing by the same way.

15) Re-assemble the pulley by reverse procedure.

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